Advances in Retrieval Augmented Generation Systems

Abstract

This paper surveys recent advances in Retrieval Augmented Generation (RAG) systems for large language models. We analyze key components including dense retrieval, chunk optimization, reranking strategies, and citation grounding. Our evaluation across five benchmark datasets shows that hybrid retrieval with reranking improves answer accuracy by 34% compared to naive RAG baselines.

1. Introduction

Large language models have demonstrated remarkable capabilities in text generation but suffer from hallucination and knowledge cutoff limitations. RAG systems address these issues by grounding model outputs in retrieved document evidence. The core pipeline consists of document ingestion, chunk indexing, query-time retrieval, and answer synthesis with source attribution.

2. Retrieval Methods

Sparse retrieval methods such as BM25 rely on keyword matching and are highly efficient but miss semantic relationships. Dense retrieval using bi-encoder models like sentence-transformers captures semantic similarity but requires more compute. Hybrid retrieval combining both approaches consistently outperforms either alone.

3. Chunking Strategies

Optimal chunk size depends on document type and query complexity. Our experiments show that 400-600 word chunks with 10% overlap yield the best retrieval precision for technical documents. Sentence-boundary-aware chunking reduces context fragmentation significantly.

4. Citation Grounding

Grounded citation generation requires the model to attribute every claim to a specific source document and page. Fine-tuned models show 89% citation accuracy versus 61% for prompted-only approaches. Inline citation formats outperform footnote styles in user comprehension studies.

5. Conclusion

RAG systems continue to evolve rapidly. Future work should focus on multi-modal retrieval, real-time index updates, and improved faithfulness evaluation metrics.